

## First steps in R

### 1 Introduction

The software used in the GLAM course is R. The goal of this first homework is to get used to the basics of the R environment and programming language.

You are expected to search information by yourself, in particular with the short R introduction available from the menu of *RGui* : Help -> Manuals (in PDF) -> An Introduction to R; or from the R-project website (<http://www.r-project.org/>). See also the icebreakeR document available on the course website.

If you have never used R before, please go through the homemade introduction to R available on the course website.

Please note that we strongly suggest that you use the RStudio software: <https://rstudio.com/products/rstudio/>. Know that if you decide to use it, you still need to download R.

### 2 Exercises

In a new script file, execute the following steps.

- a) Search information about the `read.table()` function by means of the Introduction to R, the `help(read.table)` command or `?read.table`. Execute `help.start()`, select the link “Search Engine and Keywords” and write in the box “generalized linear model” to see what happens.
- b) Save the files `BostonHousing.txt`, `breast-cancer-wisconsin.txt` and `CARDIA.dat` from the course website in your personal workspace and then import them with the `read.table()` function. Make sure you have got the whole sample and the right variables.
- c) Look for documentation about
  - graphic tools : plot, boxplot, histogram, Q-Q plot, etc.

- summary statistics : mean, standard deviation, first, second and third quartile, etc.
- d) Use the functions you found in the previous question to perform an exploratory analysis on the first two imported data sets, i.e. Example 1 and 2 presented in the first course.  
Pay attention to the fact that the response variable of the breast cancer example is coded 2 and 4, whereas the `glm()` function needs a 0/1 encoding. Another way to handle the response 2/4 is to declare it as a factor. See `help(as.factor)`.
- e) Reproduce the estimations and graphic diagnostics of both examples.  
Note that for using the `gam()` function, you have to load the `mgcv` library through Packages -> Load package... before doing any estimation <sup>1</sup>.

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<sup>1</sup>A good documentation about packages can be found on the Swiss CRAN website <http://stat.ethz.ch/CRAN/>, in the Packages section.